# AMAN – Sprint 0 POC Results (Android)

### POC 1: Values-based Blur · POC 2: Core Protection and Anti-Bypass Engine

Version 2

> **Result:** POC 1 – **Go with limitation** | POC 2 – **Go**
> Tested on a Google Pixel 4 with representative scenarios; private-tab handling was also tested on a realme (Android 16). Testing on other brands (Samsung, Xiaomi and others) belongs to the full development phase. **"Alert only"** means we detect it and alert the parent, but Android does not let us stop it.
> NEW and UPDATED mark what changed since version 1. The changes are summarised in **What changed since version 1** at the end.

## POC 2 – Core Protection and Anti-Bypass Engine

| Area | What the app does | Limitation |
|---|---|---|
| **Harmful website blocking** UPDATED | Blocks harmful sites in every browser and app, on Wi-Fi and mobile data, using a filter that runs on the phone (local VPN). Ready-made categories (Adult, Gambling, Violence, Scams, Dating, Social, Streaming). Closes searches for restricted words typed into a browser. **Now:** the Adult category uses a real list of about **64,000 adult sites** from public sources, plus a check that catches new sites whose name is clearly adult, including Arabic, Russian, Chinese, Korean and Thai words. Violence uses a small gore-site list. The larger list is looked up in a way that keeps each check fast, so browsing is not slowed down. | Blocks whole websites, not single pages. A site opened by its number address (IP) or through a web proxy is not blocked. **Now:** about **72%** of popular adult sites are caught (was under 1%). Little-known and brand-new sites are often still missed. The other categories (Gambling, Scams, Dating, Social, Streaming) are still demo lists, so production still needs a licensed list with Arabic coverage. |
| **Safe Search** | Forced on for Google (all country sites), Bing, DuckDuckGo and YouTube (app and website), in every browser including incognito. The child cannot switch it off. | Works only while our filter is running. Smaller search engines are not covered. |
| **Block-list / Allow-list** | The parent adds or removes sites, with no size limit. An "allow-list only" mode opens only the listed sites while updates, notifications and sign-in keep working. Lists are saved on the phone. A site the parent allows by name is always allowed, even if it is on the built-in adult list. | Some allowed sites may be missing pictures until the server those pictures come from is also allowed. Clearing app data erases the lists (a backend would fix this). |
| **VPN detection / restriction** | Finds VPN apps and closes them when they open. Any other VPN that connects is detected within seconds and switched off, because Android allows only one VPN and we take the slot back. The parent is alerted. Banking and government ID apps (Absher, UAE Pass and others) are kept outside the filter so they keep working. | **Alert only:** another VPN set to "Always-on" in Settings, or the child removing our VPN permission. |
| **DNS / network bypass** | Changing the phone's DNS, or turning on Chrome's "Secure DNS", does not bypass blocking. Android's Private DNS setting is checked every 2 seconds. The parent can choose VPN filter, Private DNS or Hybrid mode (Hybrid is recommended). | **Alert only:** the child changing Private DNS in Settings. **Not blocked:** little-known private DNS servers, and devices connected to the child's hotspot. |

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| **Internet Pause / Resume** | One tap pauses or resumes the internet on the whole phone (Wi-Fi and mobile data). The child sees a clear “paused by parent” notice. The pause stays in place through restarts and app kills. Calls and SMS still work. | Needs the VPN permission. **Alert only:** the child removing that permission ends the pause. |
| **Restart and tampering** | Protection restarts after a reboot. The dashboard shows health (Active / Degraded / Not running / Tampered) with how to fix it. Turning permissions off is detected and the parent is alerted. Uninstalling needs an extra step. | **Alert only:** force stop, turning permissions off, uninstalling, battery-saving and brand-specific app killers. **Not possible:** Safe Mode and “Clear data”.<br>NEW There is no parent PIN on the phone yet, so a child with the phone can change the app’s own settings (see Other findings). |

> **Decision needed – Device Owner mode.** If the app is set up during a factory reset (by QR code), most “Alert only” items become **fully prevented**: VPN and Private DNS locked, uninstall blocked, Safe Mode and hotspot disabled. The cost is that the child’s phone must be reset. **Our recommendation:** launch as a normal Play Store app and offer Device Owner later as an optional “maximum protection” mode.

## POC 1 – Values-based Blur

| Area | What the app does | Limitation |
|---|---|---|
| **Blur inside other apps** | Checks the screen about twice a second (YouTube, Instagram, TikTok, browsers). Only the bad picture is blurred, not the whole screen. The blur is real, the page can still be scrolled and tapped, and covers move with the page. Everything happens on the phone: nothing is uploaded or saved. | Android asks for screen-capture permission **again every session**, so it cannot run fully unattended (the parent gets a one-tap restart notice). There is a short delay (under about 1 second) before covering. Apps that block screenshots (banking, Netflix) cannot be checked. It uses a lot of battery. |
| **Private and incognito tabs** NEW | A private or incognito tab blocks screenshots, so before this change nothing in it could be checked and nothing was covered. Now the app recognises when a browser is showing a private tab and does what the parent chose:<br>• **Cover every picture** (default): every photo on the page is covered and the text stays readable. Covers follow the page as it scrolls.<br>• **Don’t allow private tabs:** the child is sent to the home screen with a short notice.<br>Going back to a private tab is covered again almost instantly (about 0.05 seconds). The parent’s log records only the browser and how many pictures were covered, never the page. | Pictures in a private tab cannot be judged, so all of them are covered, safe or not. Covers are plain panels rather than blur, because there is no picture to blur. Needs the Accessibility permission and a running screen-capture session. A picture that loads mid-scroll can show for a moment before its cover arrives. Banking and streaming apps are still not checked, on purpose. |
| **Parent controls** UPDATED | On/off, categories (Explicit, Suggestive, Gambling, Alcohol & smoking), strictness, blur strength, and mode (bad pictures / whole screen / cover all people). **Now also:** what to do about private tabs (cover every picture / don’t allow). | The current model is good at suggestive pictures and weaker at explicit ones, so a purpose-built model is needed before we claim safety. **Male/female targeting is not reliable**, so we recommend “cover all people”. |

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| **Browser image search** | In Chrome, Edge, Brave and Kiwi, every photo is covered as the page opens and shown only once it is judged clean, using dedicated models for explicit and drug imagery. | Android 14+ only. Explicit and suggestive content only. Photos stay covered in incognito tabs. |
| **Aman's own browser** | Images are checked before they are shown, so a bad image never appears. Ads are removed. | Protects only browsing inside Aman's browser. |

**What we changed after testing:** whole-screen covering became per-picture covering; a dark shade became a real blur; flickering covers now move with the page; an 80%-strength cover became full strength using the Accessibility permission; **private tabs, which showed everything, now have every picture covered or are not allowed.**

## Permissions Needed

| Permission | Used for |
|---|---|
| **VPN (one-time pop-up)** | Website blocking, Safe Search, lists, VPN restriction, Internet Pause |
| **Accessibility (Settings)** | Search-word blocking, full-strength blur, browser image search, **private-tab covering** UPDATED |
| **Usage Access (Settings)** | Knowing which app is open, closing VPN apps |
| **Display over other apps** | Block screen, pause screen, blur covers |
| **Screen capture (every session)** | Blur inside other apps, recognising a private tab |
| **Notifications, Battery, Device admin** | Parent alerts, staying active, making uninstall harder |

Accessibility, VPN, Usage Access and the full app list are sensitive permissions. Google Play requires a declaration for each, and parental control is an accepted use.

## What changed since version 1

| Change | Before | Now |
|---|---|---|
| **Adult site coverage** | A demo list of a few dozen sites. Measured against the adult sites people actually visit, it caught about 0.5%. | About 64,000 sites from public lists (MIT and CC BY 4.0, attributed in the app), plus a name check for new sites. Catches about **72%** of popular adult sites. |
| **Protection against over-blocking** | Not needed with a demo list. | The list was checked against the 10,000 most-visited sites. Entries that would block a shared server, a newspaper, a search engine and similar were removed, and phone services (updates, sign-in, notifications) can never be blocked by it. The name check leaves out help and recovery sites. The parent can still allow or block any site by name. |
| **Private and incognito tabs** | Could not be captured, so blur inside other apps covered nothing in them while its switch read as on. | Recognised and handled by the parent's choice: cover every picture, or don't allow private tabs. |

## Other findings

- **No parent PIN on the phone yet.** A child holding the phone can open Aman and change its settings. A parent PIN or lock is needed before launch whatever else is decided.
- **The protections already work without a backend.** Every protection runs on the phone, so the client's "independent mode" (an older child or adult using Aman for themselves, without parental monitoring) is close to the current build. Remote parental control is the part still to build. See the separate *Product Modes brief* (28 Sep).
- **Still open from version 1:** licensed site list with Arabic coverage for the other categories, purpose-built image model, and the Device Owner decision.